

Krystian Zybała

Principal JVM Performance Engineer | Low-Latency Systems Architect

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Executive Profile

Principal JVM Performance Engineer and systems architect with 10+ years of experience designing and optimizing mission-critical financial platforms. Specialized in latency-sensitive JVM runtimes, event-sourced ledgers, CPU-bound compute engines, lock-free concurrency and allocation-aware architecture. Works from application design through JIT, memory management, operating-system behaviour and hardware interaction; combines hands-on implementation with architecture ownership, mentoring and performance training.

Selected Engineering Impact

- Cut a large payment-file workload from 30–50 seconds to about four seconds and reduced memory use from roughly 10 GB to 2.5 GB through parallel execution and allocation-focused redesign.
- Reduced production infrastructure cost by approximately 90% by replacing database-centric computation with efficient in-memory execution.
- Replaced synchronized shared infrastructure with lock-free, near-zero-allocation code, removing extreme allocation pressure from a widely used logging path.
- Led JVM investigations across payment ledgers, screening, fraud and core-banking systems; optimized x86-to-ARM concurrency and JMM behaviour.

Core Expertise

JVM runtime	HotSpot, C1/C2 JIT, inlining, escape analysis, deoptimization, safepoints, GC, Java Memory Model
Performance	Tail latency, allocation elimination, CPU profiling, memory reduction, cache-aware design, benchmarking
Concurrency	VarHandles, acquire/release, lock-free algorithms, SPSC/MPSC, false sharing, thread-per-core
Architecture	Event sourcing, ledgers, payment systems, compute engines, in-memory query paths, high-throughput messaging
Low-level systems	FFM, MemorySegment, off-heap layouts, Rust, UDP, QUIC, shared memory, binary protocols, zero-copy
Toolchain	async-profiler, JFR, Linux perf, JMH, JITWatch, Datadog Profiler, flame graphs, perfasn

Professional Experience

03.2024– Present	Principal Java Engineer	NatWest Boxed
• Own performance-sensitive design around a mission-critical event-sourced payment ledger, including balance adjustments, caching, transaction and HTTP processing paths. • Reduce latency and GC pressure by simplifying hot methods, improving JIT-friendliness and removing unnecessary object creation from high-volume execution. • Diagnose production behaviour with async-profiler, JFR, Linux perf, JITWatch, Datadog Profiler, flame graphs and targeted JMH benchmarks. • Optimize locks and shared infrastructure; replaced synchronized access in a common logging path with lock-free, near-zero-allocation concurrency. • Guide x86-to-ARM work and mentor teams on JMM guarantees, VarHandles, ring buffers, cache coherence and low-latency trade-offs.		
02.2023– 02.2024	Principal Java Performance Engineer	PagoNxt
• Designed a parallel in-memory engine for large settlement-file workloads, reducing processing from 30–50 seconds to about four seconds and memory from roughly 10 GB to 2.5 GB. • Removed allocation-heavy paths, improved cache behaviour and introduced systematic profiling and JMH-based verification. • Served as Performance Champion and influenced architecture, caching and implementation of performance-critical components.		
06.2022– 12.2022	Principal Java Performance Engineer	Oscilar
• Architected a real-time fraud compute and decision engine with Kafka as event backbone and dedicated asynchronous paths for CPU-bound work. • Designed a per-thread runtime for user-defined policy evaluation; optimized event-sourced execution and contributed an in-memory query engine with ClickHouse as secondary storage.		

Earlier Experience

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| 02.2022–
06.2022 | Principal Java Performance Engineer | Silent Eight |
| <ul style="list-style-type: none">Reduced infrastructure expenditure from roughly USD 70k to USD 7k by replacing database-centric computation with efficient in-memory execution.Improved CPU utilization and reduced scoring from minutes to approximately one or two seconds while addressing ML, CPU and PostgreSQL bottlenecks. | | |
| 11.2019–
03.2022 | CTO / Architect | i-reparo.com |
| <ul style="list-style-type: none">Defined and implemented end-to-end architecture for distributed, event-driven systems and their performance strategy. | | |
| 11.2018–
01.2022 | Senior Java Performance Engineer / Architect | Ailleron S.A. |
| <ul style="list-style-type: none">Project lead for a real-time core-banking platform; co-defined architecture, technology selection, communication model and event-sourcing strategy.Designed the account domain and history model; optimized storage, time-based computation, distributed communication and DataS-tax/Cassandra query paths.Led a team whose components consistently achieved the strongest performance results within the programme. | | |
| 11.2017–
11.2018 | Senior Java Engineer | HSBC |
| 10.2016–
12.2018 | Senior Java Engineer | Tivent.pl |
| 10.2015–
10.2017 | Software Engineer | TVN / player.pl |

Selected Systems Research

Independent R&D into a deterministic ultra-low-latency runtime, presented without product branding or dates.

- Framework-free Java and Rust hot paths using FFM, off-heap layouts, allocation-free processing and explicit state ownership.
- UDP and QUIC transport, fixed-layout binary protocols, zero-copy ingest, shared-memory IPC and cache-line-aligned ring buffers.
- Thread-per-core execution, lock-free SPSC/MPSC structures, deterministic scheduling and nanosecond-level p99/p999 measurement; databases remain outside the critical path.

Consulting, Training & Talks

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| Webosystem - Founder and Principal Performance Consultant | webosystem.pl |
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- Deliver JVM performance consulting, code and architecture audits, diagnostics and design support for latency- and scalability-sensitive systems.
 - Train engineers in JVM/GC optimization, concurrency, parallelism, benchmarking, micro-optimization and high-scalability architecture.

Talks: The Power of VarHandles; Here Comes Loom; Modern GC Algorithms; Distributed Architecture Pitfalls.

Education & Additional Information

Education: College No. III im. Władysława Broniewskiego, Mathematical and Physical Profile, 2009–2012 **Languages:** Polish - native; English - proficient
Platforms: Linux, macOS, x86, ARM

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